

BUILDVISION AI

3D Building Planner User & Platform Guide

A complete walkthrough of the real-time 3D planning workspace: floors, pillars, beams, slabs, walls, doors, windows, stairs, structural validation, AI recommendations, cost estimation, and every control on screen.

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BuildVision AI is a real-time, browser-based 3D building planning platform for houses, apartments, villas, commercial buildings and offices. Everything happens inside one always-visible 3D model — there are no separate 2D/3D editing pages to switch between. Every change you make (moving a column, adding a floor, drawing a wall) instantly updates the whole structure: connected beams stretch, slabs rebuild, and material quantities recalculate live.

What you can do

- Design a multi-storey building directly in 3D with pillars, beams, slabs, walls, doors, windows and stairs.
- Get live engineering feedback — the app checks column/beam/slab capacity and geometry sanity as you work.
- See cost, concrete and steel estimates update automatically.
- Get AI layout suggestions and structural recommendations with one-click apply.
- Undo/redo every change, and multi-select groups of pillars/beams to move them together.

Who this guide is for

This guide covers everyday usage of the planner: navigating the 3D view, placing and editing structural elements, reading validation warnings, and using the AI/cost panel. It assumes no engineering background — every safety/engineering check the app runs is explained in plain language.

Accounts & access

The marketing/landing pages (home, features, pricing) are open to everyone. The planning tools themselves require an account:

- 3D Building Planner
- AI Structural Assistant
- Cost Estimator
- Webcam Measurement Tool
- Project saving & export
- Collaboration features

Visiting any of these while signed out redirects you to Login/Sign up and brings you straight back afterward. Once logged in, your session is remembered until you sign out.

Starting a project

- From the dashboard, open an existing project/building, or start a new one — the planner opens with a demo structure pre-loaded so you always have something to explore.
- The planner URL can carry a projectId/buildingId; if present, your saved building's width, length and floor count are loaded automatically.
- If nothing is saved yet, a demo office/residential structure is generated so the workspace is never empty.

The planner is one full-screen 3D scene. A floating toolbar sits at the top, an optional left panel lists floors/layers, the Property Inspector appears bottom-right when something is selected, and the Cost & AI panel slides in from the right.

Camera controls

Action	Mouse / Trackpad	Touch
Orbit	Left-drag	One-finger drag
Pan	Right-drag (or Shift+drag)	Two-finger drag
Zoom	Scroll wheel	Pinch
Select	Left-click	Tap

Camera view presets

The toolbar's view group frames the whole building instantly:

- **3D**
default perspective / three-quarter view
- **Iso**
true isometric angle, good for measuring proportions
- **Top**
plan view, looking straight down
- **Front**
front elevation, camera at building mid-height
- **Side**
side elevation
- **Reset (circular arrow icon)**
returns to the default framing

There's also an axis gizmo in the bottom-right corner of the 3D view — click any face or axis on it to snap the camera to that view, same as the toolbar buttons.

Display toggles (toolbar)

- **Inside**
walk inside the building at eye level; use WASD to move, mouse to look around, Esc to exit
- **Cutaway**
hides the roof and floors above the active one so you can see/edit the room you're on
- **Wire**
wireframe rendering of every member
- **Rebar**
shows a schematic reinforcement cage inside pillars
- **Labels**
shows floating name tags over every pillar/beam (off by default to reduce clutter — turn on when you need it)

- **Dims**
shows dimensions on the currently selected member
- **Explode**
separates floors vertically so you can see the whole stack at once
- **Snap**
snaps dragged elements to the 0.25 m grid
- **Rot**
rotates the whole building footprint 15° at a time
- **Inspect**
opens/closes the Property Inspector panel

The left tool group switches what clicking in the 3D view does. The active tool is highlighted.

Tool	What it does
Select	Click/drag pillars, walls, stairs, doors & windows. Ctrl/# + click a pillar
Pillar	Click the ground (or double-click in Select mode) to add a column at that point.
Wall	Click a start point, then an end point, to draw a wall between them.
Door	Click on an existing wall to place a door on it.
Window	Click on an existing wall to place a window on it.
Stair	Click the ground to place a staircase; it auto-sizes to the current floor height.
Delete	Click any pillar, wall, opening or stair to remove it.

Press Esc at any time to cancel a wall/stair draft, drop out of group-select, and clear the current selection.

The floor strip (toolbar, right side) lists every floor as F1, F2, ... You can have up to 20 floors.

- **Switch floor**
click its chip — this becomes the "active" floor for placing walls/doors/stairs
- **Add floor (+ Floor)**
duplicates the structural grid & walls from the floor below onto a new floor
- **Remove floor (" ")**
removes the active floor (disabled once you're down to 1 floor)
- **All floors / Isolate**
toggle whether every floor renders at once, or only the active one

Floor height, slab thickness and wall height are set per building in the building configuration; when floor height changes, connected beams, slabs and stairs automatically recompute (stairs re-fit their step count rather than changing the floor height).

Placing & moving

- With the Pillar tool active, click anywhere on the ground/active floor plane to add a column.
- With the Select tool, click-drag a pillar to reposition it — connected beams, slabs, walls and openings update live as you drag.
- Works identically with mouse, touch (drag with one finger) and — once selected — arrow keys to nudge in small steps.

Multi-select / group editing

- Ctrl (Windows) or # (Mac) + click a pillar or beam to add it to a group (h
- Ctrl/# +click again to remove it from the group.
- Drag any grouped pillar to move the whole group together as one rigid body — every connected beam/slab updates automatically.
- Delete/Backspace removes the entire selected group at once.

Pillar properties (Property Inspector)

Selecting a pillar opens its panel with editable:

- Width, depth (length), height, shape (Square / Rectangle / Circular)
- Concrete grade, steel grade, clear cover
- Longitudinal reinforcement bars & stirrup spacing/shape/hook type
- Name, rotation, and exact X/Y location
- Foundation type (isolated, combined, strip, raft, pile) via the Site tab

The inspector never pops open automatically when you click a pillar or wall — it only opens if you toggle "Inspect" in the toolbar, so selecting/dragging never gets interrupted by a panel appearing mid-drag.

Beams

Beams are generated automatically between neighbouring pillars — you do a beam to select it (this also selects the two pillars it connects to).

- Editable: width, depth, elevation, concrete/steel grade, support condition (simply supported / continuous / cantilever), top/bottom/support reinforcement.
- If a pillar moves, its beams re-snap to the new column position and re-check their depth against the new span.
- If a beam can't safely span the distance, it's flagged (see Structural Validation, next chapter) instead of silently failing.

Slabs

One slab plate covers each floor and rebuilds automatically whenever columns or beams move.

- Editable: thickness, one-way/two-way/flat system, steel direction, finish load, waterproofing, top/bottom mesh reinforcement.
- Live checks confirm the slab is thick enough for its span and load before you build.

Walls

- Select the Wall tool, click a start point, then an end point — the wall is drawn between them on the active floor.
- With the Select tool, drag any wall to reposition it (a fatter invisible hit-box makes thin walls easy to grab with mouse or finger).
- Editable: thickness, height, material, bearing type (load-bearing / non-load-bearing), lintel depth.

Doors & windows

- Select the Door or Window tool, then click anywhere on an existing wall — the opening snaps onto that wall at the clicked point.
- With Select, click a placed door/window to edit its width, height and sill height, or to delete it.
- Doors and windows only respond to clicks/edits on the currently active floor, so you never accidentally edit the wrong storey while multiple floors are visible.

Place a stair with the Stair tool, then drag it or select it to edit its properties.

How the auto-fit works

You set a target step rise (mm) and tread (mm); the number of steps is then computed live from the current floor-to-floor height, so changing the stair's width or run never changes your floor height — the stair simply re-fits itself with more or fewer steps.

Editable properties

- Width, run (depth), X/Y position
- Stair type (straight, dog-legged, spiral), rise (mm), tread (mm)
- Waist thickness

Built-in safety checks

Check	Safe range used
Step rise	150–200 mm
Tread/run	"e 2 5 0 m m
Comfort formula (2×Rise + Tread)	550–700 mm
Minimum stair width	"e 0 . 7 5 m

A stair outside any of these ranges glows amber in 3D with a hover tooltip explaining exactly which check failed — see the next chapter.

Every edit is re-checked in real time by two engines working together: a capacity engine (is this member strong enough?) and a geometry engine (does this design even make physical sense?).

What gets checked

- Missing / floating beams — a beam with no supporting column at one or both ends
- Long beam spans exceeding the safe limit for a normal RC beam
- Duplicate or overlapping columns
- Negative or zero dimensions on any pillar, beam, slab, wall or stair
- Floating slabs with no supporting beams underneath
- Zero-length walls, or walls extending outside the building footprint
- Doors/windows that no longer reference a real wall, are wider than their wall, or extend above the ceiling
- Unsafe floor heights (outside a typical 2.4–6 m range)
- Unsafe stair geometry (see previous chapter)
- Column/beam/slab capacity vs. estimated load, footing pressure vs. soil bearing capacity, and steel congestion

How issues are shown

- In 3D: the flagged member glows red (critical) or amber (warning), with a small "!" badge floating above it — hover the badge to read the explanation.
- In the toolbar: a red/amber counter next to Undo/Redo shows how many open issues exist.
- In the Cost & AI panel: every issue is listed with severity, a plain-language explanation, and — where possible — a suggested fix.

Click any issue in the panel's list to jump straight to that member in 3D (switching floor automatically if needed) and select it.

Applying a fix

Some issues include an "Apply fix" button that immediately resizes the flagged member (e.g. deepening an undersized beam, widening a footing) to the recommended value — no manual re-entry needed.

Open the right-hand panel (toolbar or the tab on the right edge) for live costing and AI guidance.

Cost & materials

- Grand total cost, split into concrete, steel, labour and formwork
- Concrete volume, steel weight, brick count, excavation volume and more
- A full bill of quantities (BOQ) broken down by category

AI structural recommendations

Lists every open validation/engineering issue with a reason, the current value, the recommended value, and (when available) an Apply fix button.

AI layout suggestions

Alternative column-grid layouts with a trade-off summary and estimated cost — click one to apply it to the whole building instantly.

- **Undo (Ctrl/# +Z, or Back button)**
step back one edit
- **Redo (Ctrl/# +Y, or Forward button)**
step forward again
- **Every drag, add, delete or property edit**
is recorded as one history step, so undo always restores exactly what you'd expect

Shortcut	Action
Ctrl/# + Z	Undo
Ctrl/# + Y	Redo
Ctrl/# + click (pillar/beam)	Toggle multi-select group
Arrow keys	Nudge selected element / group
Delete / Backspace	Remove selected element or group
Esc	Cancel draft, clear selection, exit inside view
W A S D (Inside view)	Walk around at eye level

Tips

- Turn on Labels only when you need to read names at a glance — it's off by default to keep the 3D view clean.
- Use Cutaway or Isolate when editing a specific floor in a tall building so upper floors don't get in the way.
- Use the amber/red alert counter as your first stop after a big edit — it tells you at a glance whether anything needs attention.

Known limitations (roadmap)

This is an actively evolving platform. At the time of writing:

- Stair directions are currently straight / dog-legged / spiral; U-shape, L-shape and curved stairs are planned.
- Beams don't yet have an explicit Main/Secondary/Cantilever/Transfer type field.
- Slabs don't yet expose slope, drain direction, balcony or cut-out editing in the UI.
- Center of gravity and load-path preview visualizations are planned but not yet shown.
- Large buildings aren't yet optimized with instancing/LOD — very large models may slow down.
- The Property Inspector doesn't yet have a dedicated visual history/checkpoint timeline (undo/redo works via the linear stack described above).